Topic review:

Non-tuberculous mycobacterium (NTM): rapidly growing species

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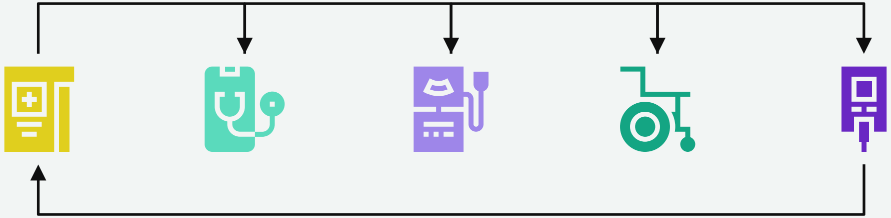
Outlines

- Introduction for NTM
- Epidemiology of NTM
- Clinical manifestations of NTM
- Laboratory diagnosis of NTM
- Treatment of NTM



01

Introduction for NTM

Nontuberculous Mycobacteria (NTM)

- Composed of species other than *M. tuberculosis* complex
- Close to 200 species of nontuberculous mycobacteria
- Only a few cause most human infections
- Previously known as ‘atypical mycobacteria’ or ‘mycobacteria other than
- *M. tuberculosis*’
- Humans are thought to acquire these organisms from environmental sources.
- There is no evidence of human-to-human transmission.

Feigin, et al. Feigin and Cherry's Textbook of Pediatric Infectious Diseases. 8th ed, W.B. Saunders,2019,p.957-1429

Nontuberculous Mycobacteria (NTM)

- NTM species are ubiquitous in soil, water, foodstuffs, and domestic and wild animals.
- **Fresh and brackish waters in warm climates, some animals** (particularly swine)
    - Human strains of *Mycobacterium avium* and *M. intracellulare* (*M. avium* complex or MAC)
- **Tap water**
    - *M. kansasii, M. xenopi, M. gordonae, M. simiae,* and *M. mucogenicum, M. fortuitum, M. chelonae,* and *M. abscessus*
- **Fish and fishtank**
    - *M. marinum*

Sarah S Long. Principles and practice of pediatric infectious dieseases.7th ed

Human pathogenic mycobacteria

- Small, aerobic rod-shaped bacilli, classified into 3 main groups

Mycobacterium tuberculosis complex

- *M. tuberculosis*
- *M. bovis*
- *M. africanum*
- *M. microti*
- *M. canetti*

M. leprae* and *M. lepromatosis

Nontuberculous mycobacteria (NTM)

- **Rapid growing mycobacteria (RGM)**
- **Slow growing mycobacteria (SGM)**

Nontuberculous mycobacteria (NTM)

Nontuberculous mycobacteria (NTM)

Slow growing mycobacteria (SGM)
(Growth within several weeks)

Photochromogen
M. kansasii
M. marinum
M. asiaticum
M. simiae

Group 1

Scotochromogen
M. szulgai
M. scrofulaceum
M. terrae
M. gordonae

Group 2

Nonchromogen
- MAC; *M. avium*, *M. intracellulare*, *M. chimaera*
- *M. haemophilum*
- *M. ulcerans**
- *M. xenopi*

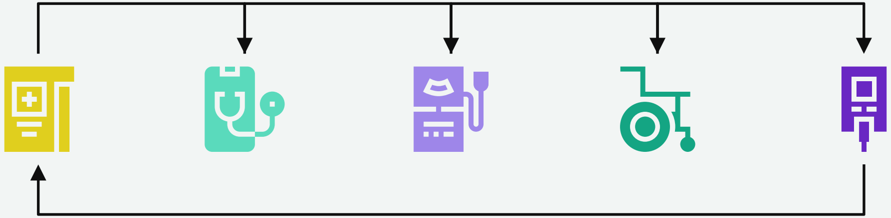
Group 3

Rapid growing mycobacteria (RGM):
(Growth within 7 days)

M. abscessus group
- *M. abscessus*
- *M. Bolletii*
- *M. massiliense*
M. fortuitum group
M. chelonae
M. smegmatis

Group 4

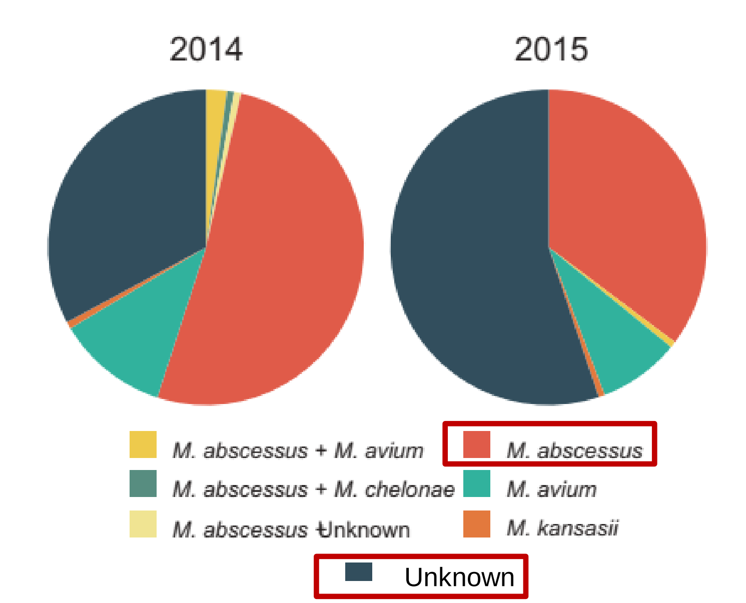
Koh W-J. Nontuberculous mycobacteria—overview. Microbiol Spectrum 5(1):TNMI7-0024-2016. 2017
Feigin, et al. Feigin and Cherry's Textbook of Pediatric Infectious Diseases. 8th ed, W.B. Saunders,2019,p.957-1429.



02

Epidemiology of NTM

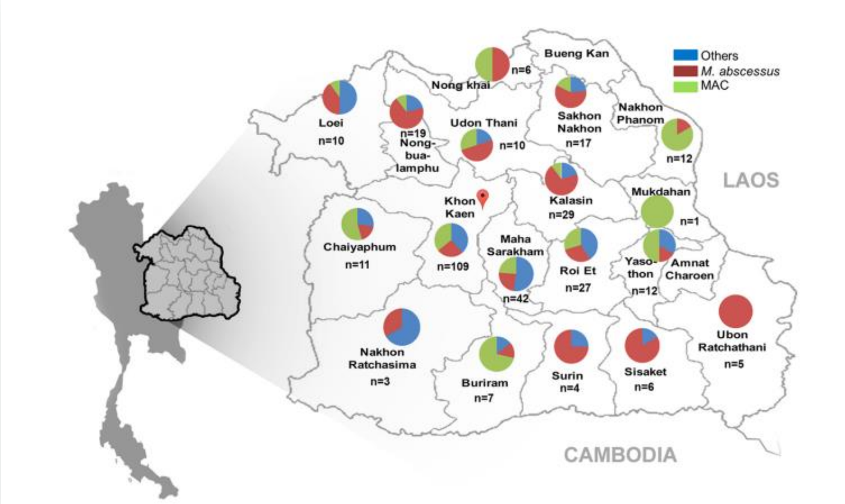
Epidemiology of NTM in children and young people with cystic fibrosis (UK)



Aaron I Gardner, et al. Clinical Infectious Diseases, Volume 68, Issue 5, 1 March 2019

Geographical distribution of NTMs causing NTM infection (Adults)

- Srinagarind Hospital, a tertiary university hospital, Khon Kaen Province, Thailand
- Clinical samples (n = 780) from 530 patients yielded NTMs in laboratory culture during the years 2012–2016



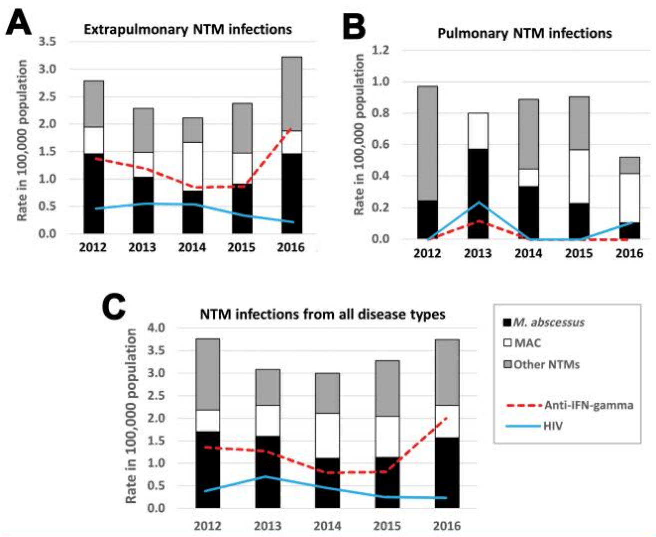
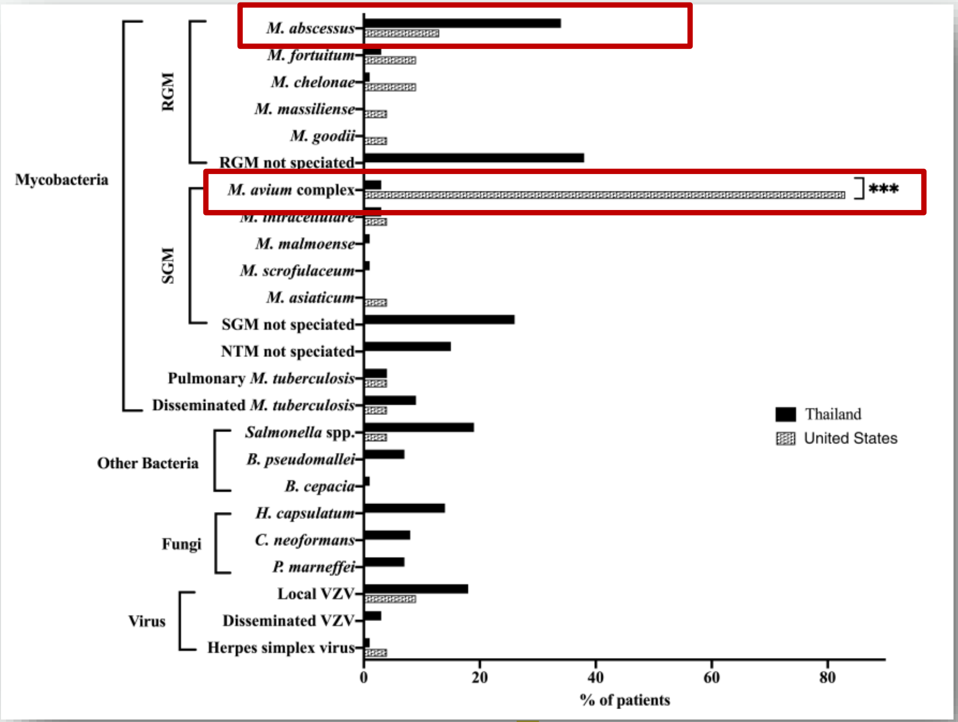


Figure 3 **The 5-year trend in NTM infection cases between 2012 and 2016 in Northeast Thailand.** The 5-year trend in the prevalence (per 100,000 hospital patient population) of extrapulmonary NTM infection (A), pulmonary NTM infection (B) and all types of NTM infection (C). All true NTM infections (*n* = 150) are plotted proportional to the number of patients in Srinagarind Hospital in the relevant year. The prevalence of anti-IFN-γ autoantibodies and HIV infection occurring in NTM infected cases are also shown. The total numbers of patients (including outpatients) visiting Srinagarind Hospital in the years 2012–2016 were 823,571, 874,195, 899,747, 883,075 and 960,664 cases, respectively.

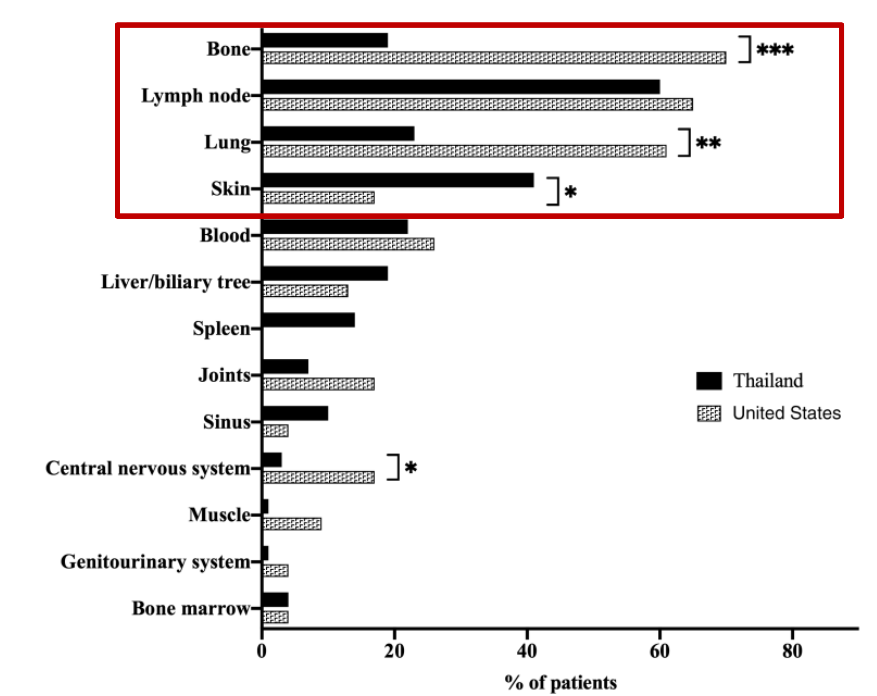
Kham-Ngam I, et al. Epidemiology of and risk factors for extrapulmonary nontuberculous mycobacterial infections in Northeast Thailand. PeerJ. 2018

Isolated organisms at presentation in Thailand and the United States



Gloria H Hong, et al. Natural History and Evolution of Anti-Interferon-γ Autoantibody-Associated Immunodeficiency Syndrome in Thailand and the United States. CID. 2020

Isolated organisms at presentation in Thailand and the United States



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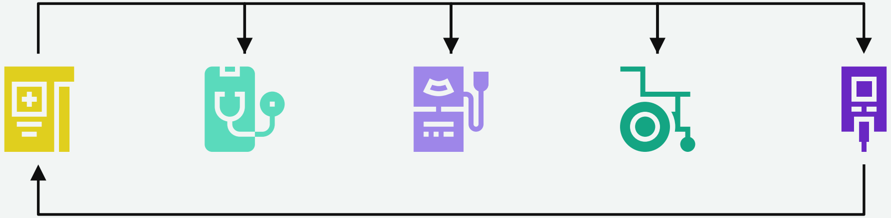
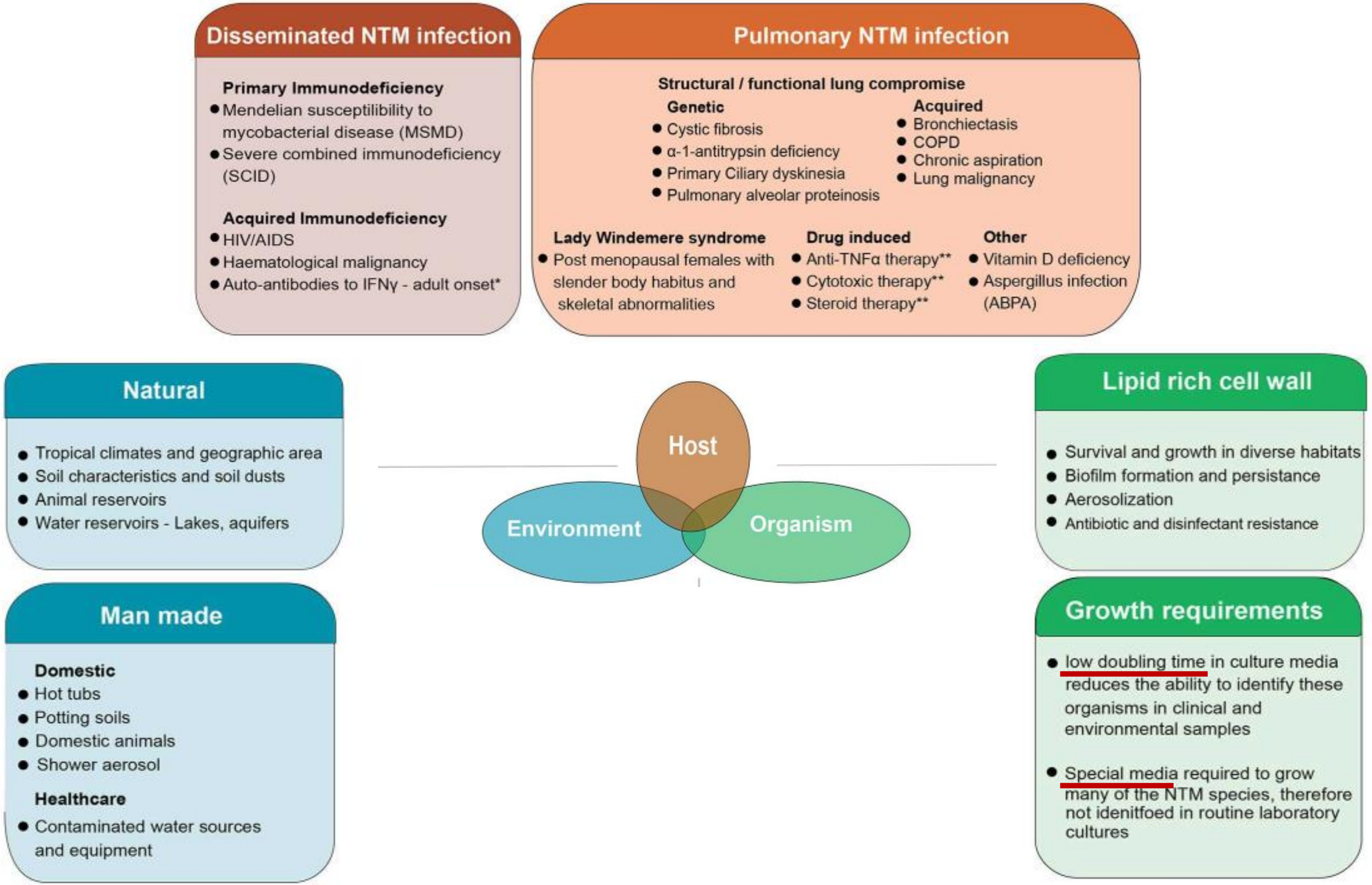


03

Clinical manifestations of NTM



Ratnatunga CN, et al. The Rise of Non-Tuberculosis Mycobacterial Lung Disease. Front. Immunol. 2020

Clinical manifestations

Clinical features

- Lymphadenitis
- Disseminated infection
- Pulmonary disease
- Skin and soft tissue disease
- Skeletal disease (bone, joint, tendon)
- Foreign body-related infection

Host factors

Impaired local immunity

- Structural lung disease, bronchiectasis, cystic fibrosis, prior TB
- Skin and soft tissue breakdown

General immunosuppression

- AIDs
- Hematologic malignancy
- Organ transplant
- Immunosuppressive therapy including steroid
- IFN-gamma: mutation, autoantibody

NTM Lymphadenitis in children

- The most common NTM clinical syndrome in children, particularly those aged 1–5 years.
- **Most common sites:**
  - Cervical and submandibular lymph nodes
- Infection usually follows peroral inoculation/ direct inoculation through the skin
- Infection is usually indolent, developing over weeks, involving unilateral lymph nodes that are non-tender, with overlying violaceous skin.
- **Common NTM species:**
  - Slow growing mycobacterium → *Mycobacterium avium complex (MAC)*
- **Treatment:** unclear
  - surgical intervention
  - antimicrobial therapy
  - a combination of these modalities or observation
- A recent meta-analysis for cure rates:
  - complete excision (98%) > antimicrobial therapy (73.1%) > observation only (70.4%)

Ford, T.J., et al. Overview of nontuberculous mycobacterial disease in children. J Paediatr Child Health. 2021

Cutaneous NTM disease in children

- Cutaneous NTM disease can present as cellulitis, non-healing ulcers, subacute or chronic nodular lesions, and abscesses.
- Most often the result of direct inoculation following disruption to the skin barrier, following soil- or water-contaminated traumatic wounds, surgeries, or cosmetic procedures
- Most affect previously healthy people
- **Most common NTM species**
  - Rapid growing mycobacterium → *Mycobacterium abscessus complex, Mycobacterium fortuitum* and *Mycobacterium chelonae*
- **Treatment:**
  - Depending on the organism identified
  - Often resolve without therapy for Isolated, uncomplicated cutaneous NTM lesions
  - Antimycobacterial therapy and/or surgical management

Redbook, 32nd edition
Ng, S.S.-Y., et al. Pediatric Cutaneous Nontuberculous Mycobacterium Infections in Singapore. Pediatr Dermatol. 2015
Ford, T.J., et al. Overview of nontuberculous mycobacterial disease in children. J Paediatr Child Health. 2021

NTM pulmonary disease in adults

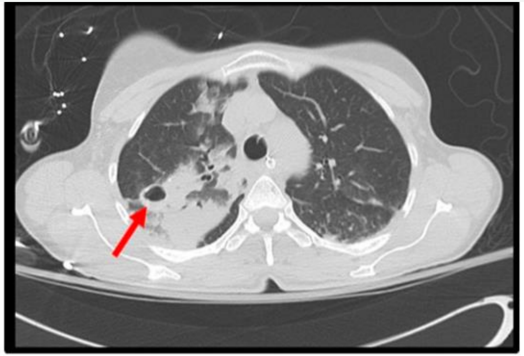
- **Common etiologic agents**
  - Slow grower: MAC, *M. kansasii, M. malmoense, M. xenopi*
  - Rapid grower: *M. abscessus*
- **Clinical and Microbiologic Criteria for Diagnosis of Nontuberculous Mycobacterial Pulmonary Disease (Adult)**

Clinical	Pulmonary or Systemic Symptoms	Both Required
Radiologic	Nodular or cavitary opacities on chest radiograph, or a high-resolution computed tomography scan that shows bronchiectasis with multiple small nodules	
and	Appropriate exclusion of other diagnoses	
Microbiologic[b]	1. Positive culture results from at least two separate expectorated sputum samples. If the results are nondiagnostic, consider repeat sputum AFB smears and cultures **or** 2. Positive culture results from at least one bronchial wash or lavage **or** 3. Transbronchial or other lung biopsy with mycobacterial histologic features (granulomatous inflammation or AFB) and positive culture for NTM or biopsy showing mycobacterial histologic features (granulomatous inflammation or AFB) and one or more sputum or bronchial washings that are culture positive for NTM	

Daley CL, et al. Treatment of Nontuberculous Mycobacterial Pulmonary Disease: An Official ATS/ERS/ESCMID/IDSA Clinical Practice Guideline. CID 2020

NTM Pulmonary disease in children

- Associated with underlying pulmonary pathology, most commonly cystic fibrosis (CF)-bronchiectasis
- **Clinical presentation:**
  - Non-specific
  - Worsening pre-existing symptoms, such as cough or sputum production, decline in underlying lung function
  - Constitutional symptoms, such as fever and weight loss.
- **Radiological findings:**
  - Nodular or cavitary opacities, or bronchiectasis
- Pulmonary NTM diagnostic criteria have been developed, however these have not been validated clinically.



Ford, T.J., et al. Overview of nontuberculous mycobacterial disease in children. J Paediatr Child Health. 2021

Hendrix, C, et al. ; Diagnosis and Management of Pulmonary NTM with a Focus on *Mycobacterium avium* Complex and *Mycobacterium abscessus*: Challenges and Prospects. *Microorganisms* 2023

Disseminated NTM disease in children

- Disseminated NTM disease in children is uncommon, always indicating immunocompromise.
- **Most common NTM species:**
  - *Mycobacterium avium complex (MAC)*
- **Clinical features:**
  - non-specific, fever, weight loss, lymphadenopathy, skin lesions and GI symptoms
- **Disseminated NTM disease**: associated with impaired cell-mediated
  - **Congenital immune defects**
    - Interferon-gamma receptor defects
    - IL-12 deficiency
    - Cytokine-JAK-STAT pathways abnormalities
    - NF-kappa-B essential modulator [NEMO] mutation and related disorders
  - **HSCT**
  - **HIV**
- Defects in these pathways also predispose to infections with other intracellular pathogens
  - Salmonella spp., Coccidioides spp., Histoplasma spp. and various viral pathogens

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Ford, T.J., et al. Overview of nontuberculous mycobacterial disease in children. J Paediatr Child Health. 2021

Complex cellular pathways

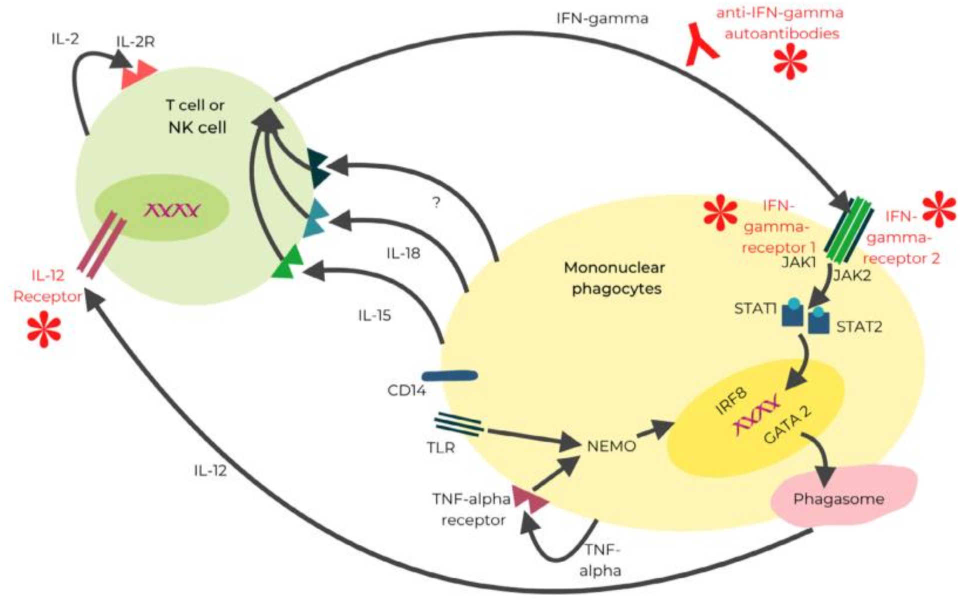

Fig 1 Complex cellular pathways of Mendelian control (Adapted from Wu et al.,[8] with permission). *Common defects in MSMD. IFN, interferon; IL, interleukin; IRF, interferon regulatory factor; JAK, Janus kinase; NEMO, nuclear factor-kB–essential modulator; STAT, signal transducer and activator of transcription; TNF, tumour necrosis factor; TLR, toll like receptor.

Ford, T.J., et al. Overview of nontuberculous mycobacterial disease in children. J Paediatr Child Health. 2021

Other clinical manifestations of NTM

- **Less common**
    - Osteomyelitis
    - Otitis media
    - Central catheter-associated blood stream infections

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Diseases caused by NTM in children

The **incubation periods** are variable.

Clinical Disease	Species
Cutaneous infection	*Mycobacterium marinum* *Mycobacterium chelonae* *Mycobacterium fortuitum* *Mycobacterium abscessus* ★ *Mycobacterium ulcerans*[a]
Lymphadenitis	MAC *Mycobacterium haemophilum* *Mycobacterium lentiflavum* *Mycobacterium kansasii* *M fortuitum* *M abscessus* ★ *Mycobacterium malmoense*[b]
Otologic infection	*M abscessus* *M fortuitum*
Pulmonary infection	MAC *M kansasii* *M abscessus* ★ *Mycobacterium xenopi* *M malmoense*[b] *Mycobacterium szulgai* *M fortuitum* *Mycobacterium simiae*
Catheter-associated infection	*M chelonae* *M fortuitum* *M abscessus* ★
Prosthetic valve endocarditis	*M chelonae* *M fortuitum* *Mycobacterium chimaera*
Skeletal infection	MAC *M kansasii* *M fortuitum* *M chelonae* *M marinum* *M abscessus* ★ *M ulcerans*[a]
Disseminated	MAC *M kansasii* *Mycobacterium genavense* *M haemophilum* *M chelonae*

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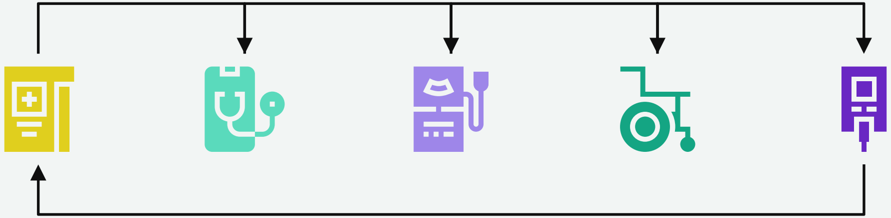


04

Laboratory diagnosis of NTM

Characteristics of Rapid growing mycobacteria (RGM)

TABLE 97.2 **Characteristics of Rapid Mycobacterial Growers**

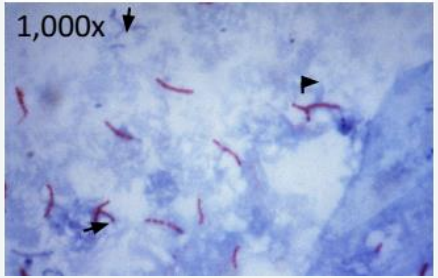
Runyon Group IV	Optimal Temperature (°C)	Growth Rate (Days)	Niacin	Nitrate Reduction	Growth in 5% NaCl
M. fortuitum	37	3–7	–	+	+
M. abscessus	37	3–7	–	–	+
M. chelonae	37	3–7	–	–	–

Feigin, et al. Feigin and Cherry's Textbook of Pediatric Infectious Diseases. 8th ed, W.B. Saunders,2019

Diagnostic tests

- **Definitive diagnosis of NTM disease**
  - requires isolation of the organism
- **How to collect specimen:**
  - isolation of *Mycobacterium haemophilum* requires that the culture be maintained at 30°C and that hemecontaining medium is added for isolation.
  - contamination of cultures or transient colonization can occur

- An acid-fast bacilli smear-positive sample and repeated isolation on culture media of a single species from any site are more likely to indicate disease than culture contamination or transient colonization.



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Diagnostic tests

- **Diagnostic criteria for NTM pulmonary disease**
    - **In adults** include 2 or more separate sputum samples or 1 bronchial alveolar lavage specimen that grows NTM.
    - These criteria have not been validated in children and apply best to MAC, *M kansasii,* and *M abscessus.*
- **Site of clinical significance for NTM:**
    - Draining sinus tracts or wounds
    - Sterile sites:
        - cerebrospinal fluid, pleural fluid, bone marrow, blood, lymph node aspirates, middle ear or mastoid aspirates, or surgically excised tissue

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Laboratory diagnosis of NTM pulmonary disease

- Obtaining respiratory samples
- Sample processing and culture
- Species identification
- Drug susceptibility testing

IDSA
Infectious Diseases Society of America

Treatment of Nontuberculous Mycobacterial Pulmonary Disease. Clinical Infectious Diseases, 2020

Laboratory diagnosis of NTM pulmonary disease

Obtaining respiratory samples

- Given the slow course of NTM pulmonary disease, a prolonged interval ensures that repeat positive cultures are unlikely to reflect a transient contamination of the tracheobronchial system after a single environmental exposure
- occasional presence of NTM in the tracheobronchial tract, at least 3 respiratory samples are investigated, over an interval of at least a week

Treatment of Nontuberculous Mycobacterial Pulmonary Disease. Clinical Infectious Diseases, 2020

Laboratory diagnosis of NTM pulmonary disease

Sample Processing and Culture

- Decontamination by 0.25% N-acetyl-L-cysteine and 1% NaOH (NALC-NaOH)
- Respiratory samples: 3 morning specimens on different days
- Should identified clinically significant NTM isolates to the species level
  - *M.a. massiliense, M.a. abscessus*
- CLSI: incubations temperatures
  - 36 ± 1 °C for slow growers
  - 28 ± 2 °C for rapid growers

Treatment of Nontuberculous Mycobacterial Pulmonary Disease. Clinical Infectious Diseases, 2020

Laboratory diagnosis of NTM pulmonary disease

- Microscopic slide examination smear
- Culture on both solid and liquid media to increase sensitivity
    - **Solid media → Löwenstein-Jensen (LJ) medium**
        - Colony morphology, detect mixed infection, detect NTM even incomplete sample decontamination
    - **Liquid media → MGIT**
        - More rapid/sensitive, better for rapid growing NTM

Treatment of Nontuberculous Mycobacterial Pulmonary Disease. Clinical Infectious Diseases, 2020

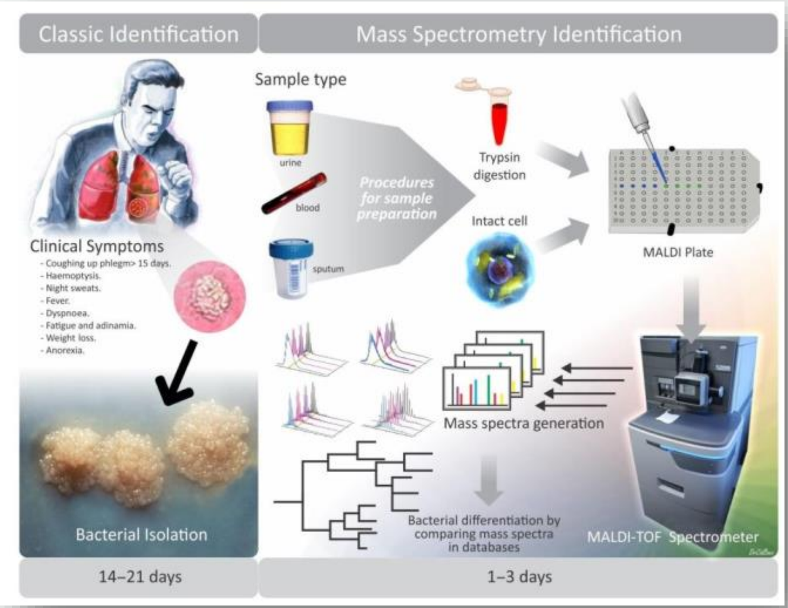
Conventional Methods for Identification

- Observation of rate of growth (> 7 days, < 7 days)
- Colony morphology
- Pigmentation (Runyon classification (1959)
    - Rapid grower → Non-chromogenic
    - Slow grower → Photochromogen , Scotochromogens, Nonchromogens
- Require 6-8 weeks

Laboratory diagnosis of NTM pulmonary disease

Species Identification (1)

- Molecular identification is the preferred method and can be achieved using probes or gene sequencing.
  - Line probe assays (LPA)
- MALDI-TOF (the matrix-assisted laser desorption ionization-time of flight mass spectrometry) method

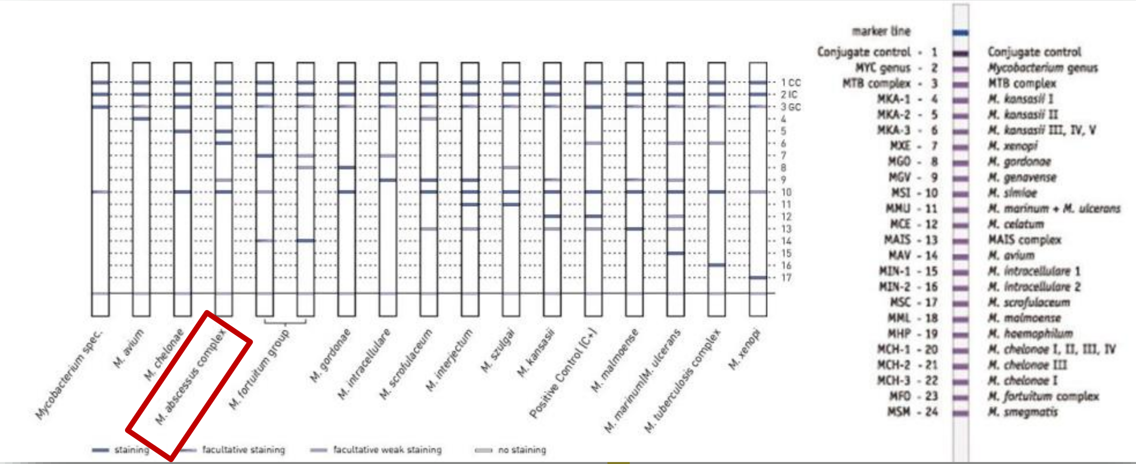


Treatment of Nontuberculous Mycobacterial Pulmonary Disease. Clinical Infectious Diseases, 2020

Laboratory diagnosis of NTM

Species Identification (2)

- **Line probe assays:**
  - **INNO-LiPA® MYCOBACTERIA**
    - Detection of MTBC and differentiation of 16 clinically relevant NTM species from cultures (16S-23S rRNA spacer region)



Laboratory diagnosis of NTM pulmonary disease

Drug Susceptibility Testing (1)

M. abscessus **pulmonary disease**

- **Parenteral drugs (in vitro activity):**
  - Amikacin, imipenem, cefoxitin, and tigecycline
- **Oral drugs (some activity):**
  - Macrolides, oxazolidinones (linezolid) and clofazimine
    - Clofazimine shows in vitro activity, acts synergistically with amikacin and macrolides and prevents the emergence of amikacin-resistant *M. abscessus* in vitro

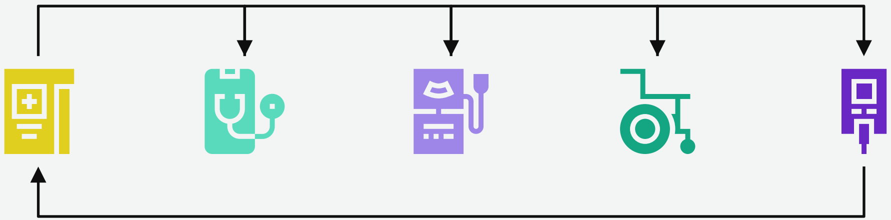
Treatment of Nontuberculous Mycobacterial Pulmonary Disease. Clinical Infectious Diseases, 2020

Laboratory diagnosis of NTM pulmonary disease

Drug Susceptibility Testing (2)

- Strains of *M. abscessus* subsp. *abscessus* and *M. abscessus* subsp. *bolletii*
  - erythromycin resistance methylase (*erm*) gene, named *erm*(41) → results in inducible resistance to macrolides
- **Susceptibility testing panels for *M. abscessus:***
  - include at least amikacin, cefoxitin, imipenem, clarithromycin, linezolid, doxycycline, tigecycline, ciprofloxacin, and moxifloxacin.
- CLSI recommends that drug susceptibility testing be performed by broth microdilution

Treatment of Nontuberculous Mycobacterial Pulmonary Disease. Clinical Infectious Diseases, 2020



05

Treatment of NTM

Treatment of NTM

- Many NTM are relatively resistant in vitro to antituberculosis drugs
- **Initial therapy should be directed by the following:**
  - ✓ Species
  - ✓ Drug-susceptibility testing, especially to macrolides
  - ✓ Site(s) of infection: localized vs disseminated
  - ✓ the patient's immune status
  - ✓ the need to treat a patient presumptively for tuberculosis while awaiting culture reports that subsequently reveal NTM

- The choice of drugs, dosages, and duration should be reviewed with a consultant experienced in the management of NTM infections, but always includes 2 or more drugs.

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Treatment of Nontuberculous Mycobacteria
(Rapidly growing species)

Organism	Disease	Redbook
Mycobacterium abscessus	• Otitis media • Cutaneous infection	• No reliable antimicrobial regimen because of variability in drug susceptibility • Clarithromycin plus initial course of amikacin plus cefoxitin or imipenem/meropenem; • may require surgical debridement on the basis of in vitro susceptibility testing (50% are amikacin resistant)
	• Pulmonary infection	• Serious disease, clarithromycin, amikacin, and cefoxitin or imipenem/meropenem on the basis of susceptibility testing; • most isolates have very low MICs to tigecycline; may require surgical resection.

Redbook, 32nd edition

Treatment of Nontuberculous Mycobacteria
(Rapidly growing species)

Organism	ATS/ERS/ESCMID/IDSA (Adults)
Mycobacterium abscessus	• Suggest susceptibility-based treatment for macrolides and amikacin over empiric therapy • **Based on pulmonary NTM** - Strains *without* inducible or mutational resistance → suggest macrolide-containing multidrug treatment regimen - Strains *with* inducible or mutational resistance → suggest macrolide- containing regimen if the drug is being used for its immunomodulatory properties although the macrolide is not counted as an active drug in the multidrug regimen • Multidrug regimen: at least 3 active drugs (guided by in vitro susceptibility) in the initial phase of treatment

Treatment of Nontuberculous Mycobacterial Pulmonary Disease. Clinical Infectious Diseases, 2020

ตารางที่ 4-1 ความไวของยากลุ่ม macrolide ต่อเชื้อ *M. abscessus* แต่ละ subspecies และตำแหน่งยีนที่เกี่ยวข้องกับการดื้อยาชนิดต่าง ๆ

Subspecies of *M. abscessus* complex	subsp. *massiliense* (some strains of subsp. *abscessus**)	• subsp. *bolletii* • subsp. *abscessus*	All subspecies
Clarithromycin ทดสอบความไวในวันที่ 3-5	ไว	ไว	ดื้อ
Clarithromycin ทดสอบความไวในวันที่ 14	ไว	ดื้อ	ดื้อ
Macrolide susceptibility phenotype	Macrolide susceptible	Inducible macrolide resistance	High-level constitutive macrolide resistance
Genetic implication	Dysfunctional erm**(41) gene	Functional erm (41) gene	23S ribosomal RNA point mutation (rrl mutation)
Macrolide effect	Anti-mycobacterial	Immuno-modulatory	Immuno-modulatory

*ร้อยละ 15-20 ของ *M. abscessus* subsp. *abscessus* อาจจะเป็น dysfunctional erm41 แต่ส่วนใหญ่ ร้อยละ 80 จะเป็น functional erm 41 gene

**erm: erythromycin resistance methylase

ตารางที่ 4-2 จำนวนและชนิดยาที่แนะนำในการรักษา *M. abscessus* ในระยะแรก และ ระยะต่อเนื่อง แบ่งตามเชื้อที่มีความไวต่อยากลุ่ม macrolide แบบต่าง ๆ

ความไวต่อยากลุ่ม macrolide		จำนวนยาที่แนะนำ	ยาที่แนะนำ (preferred drugs)
Mutational	Inducible**		
Susceptible*	Susceptible	ระยะแรก (initial phase) ≥ 3 ชนิด	ยาชนิดฉีด (เลือก 2 ชนิดอย่างน้อย 2-4 สัปดาห์แรก) Amikacin Imipenem (หรือ cefoxitin) Tigecycline ยาชนิดกิน (เลือก 1-2 ชนิด) Azithromycin (หรือ clarithromycin) Clofazimine Linezolid
		ระยะต่อเนื่อง (continuation phase) ≥ 2 ชนิด	ยาชนิดกิน (เลือก 2-3 ชนิด) Azithromycin (หรือ clarithromycin) Clofazimine Linezolid

ความไวต่อยากลุ่ม macrolide		จำนวนยาที่แนะนำ	ยาที่แนะนำ (preferred drugs)
Mutational	Inducible**		
Susceptible หรือ Resistant***	Resistant	**ระยะแรก** (initial phase) ≥ 4 ชนิด	**ยาชนิดฉีด** (เลือก 2-3 ชนิด อย่างน้อย 4 สัปดาห์แรก) Amikacin Imipenem (หรือ cefoxitin) Tigecycline **ยาชนิดกิน** (เลือก 2-3 ชนิด) Azithromycin (หรือ clarithromycin) Clofazimine Linezolid
		ระยะต่อเนื่อง (continuation phase) ≥ 3 ชนิด	**ยาชนิดกิน** (เลือก 3 ชนิด) Azithromycin (หรือ clarithromycin) Clofazimine Linezolid

*Macrolide susceptible: MIC ≤ 2 มคก./มล. ในวันที่ 3 และ วันที่ 14

**Inducible macrolide resistance: MIC ≤ 2 มคก./มล. ในวันที่ 3 แต่ ≥ 8 มคก./มล. ในวันที่ 14, ถ้ามี macrolide resistance หรือ inducible macrolide resistance ยังสามารถให้ยาได้ เพื่อหวังผลของ immuno-modulatory effect แต่จะไม่นับเป็นยาที่ไว 1 ชนิด โดยยา clarithromycin ทำให้เกิด inducible macrolide resistance ได้มากกว่า azithromycin

***Macrolide resistance: MIC ≥ 8 มคก./มล. ตั้งแต่วันที่ 3

Treatment of Nontuberculous Mycobacteria
(Rapidly growing species)

Organism	Disease	Initial Treatment
***Mycobacterium fortuitum* group**	**Cutaneous infection**	• Initial therapy for serious disease is amikacin plus meropenem IV • Followed by clarithromycin, doxycycline* or trimethoprim/sulfamethoxazole, or ciprofloxacin, orally, on the basis of in vitro susceptibility testing; may require surgical excision. • Up to 50% of isolates are resistant to cefoxitin
	Catheter infection	• Catheter removal and amikacin plus meropenem, IV; clarithromycin, trimethoprim-sulfamethoxazole, or ciprofloxacin, orally, on the basis of in vitro susceptibility testing.

*Treatment always includes 2 or more drugs
*Doxycycline can be used for short durations (ie, 21 days or less) without regard to patient age, but for longer treatment durations is not recommended for children younger than 8 years (see Tetracyclines, p 866). Only 50% of isolates of *M marinum* are susceptible to doxycycline.

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Treatment of Nontuberculous Mycobacteria
(Rapidly growing species)

Organism	Disease	Initial Treatment
Mycobacterium chelonae	Catheter infection, prosthetic valve endocarditis	• Catheter removal; debridement, • removal of foreign material; valve replacement; and • Tobramycin (initially) plus clarithromycin, meropenem, and linezolid
	Disseminated cutaneous infection	• Tobramycin and • Meropenem or linezolid (initially) plus • Clarithromycin

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Treatment regimen for the most common NTM infections

M . abscessus	• Pulmonary • Disseminated • Extensive cutaneous	Amikacin (15 mg/kg/day iv in 1or 2 doses, peaks at 20 μg/ml, if given in 2 divided doses) + clarithromycin (1g once daily or 500 mg twice daily) + cefoxitin (200 mg/kg, usually 12 g/die iv) or imipenem (500–750 mg three times daily iv)	Unknown[d]
	• Cutaneous localized	Clarithromycin (500 mg twice daily)	4–6 months
M. fortuitum	• Pulmonary (P) • Cutaneous (C)	Clarithromycin (500 mg twice daily) + doxycycline (100 mg daily) or trimethoprim-sulfamethoxazole (800+160 mg twice daily) or levofloxacin (500–750 mg daily) or moxifloxacin (400 mg daily) or gatifloxacin (400 mg daily)	6–12 months (P) 4–6 months (C)

Wagner D, Young LS. Nontuberculous mycobacterial infections: a clinical review. Infection. 2004;32(5):257-70

Treatment of NTM

- **Duration of therapy:**
  - depend on host status, site(s) of involvement, and severity
- Most patients who respond ultimately show substantial clinical improvement in the first 4 to 6 weeks of therapy.
- Elimination of the organisms from blood cultures can take longer, often up to 12 weeks.
- Recommend a minimum treatment duration of 3 to 6 months or longer.

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Table 4
***In vitro* susceptibility of selected NTM to various antimycobacterial agents.**

Antibiotic [reference]	MAC	KAN	ABS	FOR	CHE	MAL[a]	XEN[a]	MAR	HAE
Rifampin	++	++	-	-	-	-	++	++	++
Rifabutin	++	++	-	-	-	+/-	++	++	++
Isoniazid	- [b]	++	-	-	-	-	++	-	
Ethambutol	++	++	-	-	-	++	++	+/- [c]	-
Pyrazinamide	-	-	-	-	-			-	
Ethionamide	++	++		-	-	++			
Streptomycin	++	++	-	-	-	-	+	+/-	
Kanamycin	++			+/-	+/-	++			+/-
Amikacin	++	++	+	++	++	++	++	++	++
Tobramycin				+/-	++				
Cycloserine	++			-	-	++			++
PAS	-			-	-	-			
Capreomycin	-	-			-	-	-		
Doxycycline [35]	-	+/-	-	+/-	-			+	+/-
Minocycline [35]	-	++	-	+	+/-			++	
Tigecycline [35]	-	-	++	++	++			-	
Ciprofloxacin	++	++	-	++	-	+/-	++	-	++
Levofloxacin [81]	++	++	-	++	-	++	++	+/-	++
Moxifloxacin [82, 83]	++	++	-	++	-		++	++	++
Clarithromycin	++	++	++	+	++	++	++	++	++
TMP-SMZ	-	+	-	++	-			++	+/-
Cefoxitine	-		+	+	-				
Clofazimine	+/-		++		-				
Imipenem			+/-	++	+/-			++	
Linezolid [84, 85]	+/-	++	+/-	++	++	++	++	++	
Mefloquine [86]	++								

Symbols represent approximate susceptibility ranges: ++: 90–100%; +: 60–90%; +/-: 30–60%; - : 0–30%. MAC: *M. avium* complex [87]; KAN: *M. kansasii* [34]; ABS: *M. abscessus* [19 , 51, 60]; FOR: *M. fortuitum* [20, 52, 61, 87]; CHE: *M. chelonae* [20, 52, 61, 87]; MAL: *M. malmoense* [2, 88]; XEN: *M. xenopi* [2]; MAR: *M. marinum* [89, 90]; HAE: *M. haemophilum* [2]; [a] lack of consistency in susceptibility results [2]. [b] see [32] for further discussion; [c] in [89] discussed as resistant

In vitro susceptibility of selected NTM to various antimycobacterial agents

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Dosage of antimycobacterial Agents

TABLE 97.3 Antimycobacterial Agents

Drug	Dosage	Form
Amikacin (A)	15–30 mg/kg/day divided 3 times daily	IV or IM
Azithromycin (Z)	500 mg bid (adults/adolescents); 10–12 mg/kg/day (children)	PO
Cefoxitin (X)	80–160 mg/kg/day divided 4 to 6 times daily	IV or IM
Ciprofloxacin (C)	20–30 mg/kg/day divided 2 times daily (adults only in United States)	PO or IV
Clarithromycin (CL)	15–30 mg/kg/day divided 2 times daily	PO
Clofazimine (CLO)	1–2 mg/kg/day	PO
Doxycycline (D)	2–4 mg/kg/day divided 2 times daily (>8 yr old)	PO, IV
Ethambutol (ETB)	15 mg/kg/day	PO
Ethionamide (ETH)	15–20 mg/kg/day divided 2 times daily	PO
Imipenem (IMP)	60–100 mg/kg/day divided 4 times daily	IV
Isoniazid (I)	5 mg/kg/day	PO
Pyrazinamide (PZA)	15–30 mg/kg/day	PO
Rifabutin (RIB)	5–10 mg/kg/day; maximum 300 mg/day	PO
Rifampin (RIF)	10–20 mg/kg/day once or divided 2 times daily	PO or IV
Streptomycin (S)	20–30 mg/kg/day	IM

INFECTIONS

Disseminated MAC: HIV infected: CL (or Z) + ETB (± RIB)

MAC lung disease: CL (or Z) + ETB + RIF + S or A (severe or cavitary disease)

Mycobacterium abscessus: CL or Z + one or more of A, X, or IMP

Mycobacterium chelonae: CL alone (skin disease), two or more of the following: CL, linezolid, IMP, or tobramycin (invasive disease)

Mycobacterium fortuitum: two or more of the following: CL, C, A, X, IMP, or TMP-SMX

Mycobacterium kansasii: RIF + ETB + I

Mycobacterium marinum: ETB + RIF, or CL + RIF

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Isolation of the hospitalized patient

- Standard precautions are recommended.

Take home messages

- NTM → found in environmental sources
- **Rapid grower NTM:**
  - *M. abscessus group, M. fortuitum group, M. chelonae*
- Cervical lymphadenopathy is a common NTM infection in children
- Disseminated NTM disease in children → work up immunodeficiency syndrome e.g. IFN-gamma mutation
- **Diagnosis and distinction of NTM**
  - need a high index of suspicion, and involves a variety of diagnostic modalities (tissue patho, AFB smear, culture, molecular, imaging)
- **Treatment of NTM infection**
  - depends on the strain and anatomical site of infection, involves antibiotic combinations, surgery, or both.
  - Usually use multidrug regimen
- Drug resistant NTM is a challenging problem.

Thank you